

PORTERVILLE, CA, USA

WMO: 723895

Lat: 36.029N

Lon: 119.063W

Elev: 135

StdP: 99.72

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23149

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.0	0.6	-6.2	2.3	4.9	-4.0	2.7	6.3	6.9	11.3	5.5	11.3	1.4	110	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	38.7	21.6	37.5	20.8	36.3	20.2	22.9	36.1	21.9	34.8	21.2	34.1	4.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.1	30.2	17.2	12.5	29.5	16.1	11.7	28.5	68.1	35.9	64.5	35.0	61.6	33.5	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.8	5.1	4.7	-2.5	41.3	1.4	1.2	-3.6	42.2	-4.4	42.9	-5.2	43.5	-6.2	44.3		
(5)			-3.3	24.0	1.1	1.3	-4.1	25.0	-4.8	25.7	-5.4	26.4	-6.2	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	8.8	10.5	13.1	15.7	19.9	24.1	27.2	26.3	23.7	18.0	12.2	8.5		
	DBStd	7.29	2.76	2.73	3.18	3.82	3.84	3.63	2.67	2.52	3.26	3.39	3.13	2.87		
	HDD10.0	172	58	24	8	2	0	0	0	0	0	0	14	66		
	HDD18.3	1346	297	219	164	97	27	3	0	0	2	46	184	306		
	CDD10.0	2860	19	38	104	174	306	424	532	504	411	249	80	18		
	CDD18.3	992	0	0	2	18	74	177	274	246	163	37	1	0		
	CDH23.3	14158	0	3	78	373	1178	2612	3908	3332	2088	557	29	0		
	CDH26.7	7844	0	0	13	126	545	1489	2381	1977	1126	186	2	0		
(14) Wind	WSAvg	2.2	1.7	1.9	2.2	2.5	2.7	2.8	2.7	2.5	2.3	2.1	1.7	1.7		
(15) Precipitation	PrecAvg	273	47	47	46	28	9	2	0	1	5	16	39	37		
	PrecMax	494	201	166	135	121	74	12	4	7	39	65	89	142		
	PrecMin	133	1	2	0	0	0	0	0	0	0	0	0	0		
	PrecStd	91	42	44	32	29	16	3	1	1	10	19	26	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	23.1	28.0	32.6	37.3	40.2	41.0	39.9	37.9	33.8	26.6	20.1		
		MCWB	13.3	14.8	16.8	17.8	19.9	22.2	22.8	22.3	21.0	18.9	15.6	13.0		
	2%	DB	17.6	20.9	25.9	30.1	34.1	37.9	38.8	37.8	36.4	31.1	23.7	17.7		
		MCWB	12.1	14.0	15.9	16.9	18.5	20.4	21.9	21.0	20.0	17.8	14.7	11.9		
	5%	DB	16.0	18.0	22.9	27.6	32.3	36.3	37.6	37.0	34.8	28.6	21.4	16.0		
		MCWB	11.7	12.7	14.7	16.0	17.7	19.7	21.3	20.7	19.6	16.7	13.7	11.2		
	10%	DB	13.9	16.4	21.2	25.0	29.7	34.1	36.3	35.3	32.8	27.0	18.8	13.9		
		MCWB	10.7	11.7	13.8	15.0	17.1	19.0	20.6	20.0	18.8	16.0	12.7	10.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	16.0	18.2	19.9	21.1	23.7	24.3	23.5	23.0	20.2	17.0	14.9		
		MCDB	17.8	20.3	25.5	29.2	34.8	37.4	37.5	37.4	35.5	30.8	24.1	18.1		
	2%	WB	13.3	14.8	16.7	17.8	19.6	21.8	23.1	22.3	21.2	18.8	15.6	13.3		
		MCDB	16.1	19.2	24.3	28.0	32.8	35.5	36.3	35.4	33.4	28.4	21.9	15.9		
	5%	WB	12.2	13.5	15.4	16.5	18.5	20.7	22.2	21.4	20.2	17.7	14.5	12.0		
		MCDB	15.0	17.5	22.1	26.3	30.5	34.4	35.1	34.4	32.9	26.7	20.0	14.7		
	10%	WB	11.1	12.3	14.1	15.4	17.4	19.7	21.3	20.7	19.4	16.5	13.3	10.8		
		MCDB	13.7	15.9	19.9	24.2	28.7	33.0	34.2	33.7	32.0	25.7	18.1	13.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	11.3	13.8	15.3	16.3	17.8	17.4	17.6	17.1	15.2	12.6	10.4		
		MCDBR	12.9	14.2	17.3	18.9	19.1	19.3	17.9	18.3	18.4	18.1	16.4	13.1		
		MCWBR	8.7	9.0	9.3	8.7	7.8	7.7	7.0	7.3	7.7	8.3	9.4	8.5		
	5% WB	MCDBR	10.9	12.8	16.1	17.5	18.2	18.4	17.1	17.3	17.4	15.7	14.2	10.0		
		MCWBR	7.6	8.6	9.1	8.5	7.7	7.5	6.9	7.2	7.5	7.4	8.5	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.364	0.365	0.396	0.404	0.388	0.422	0.397	0.388	0.391	0.359	0.354		
	taud		2.344	2.321	2.335	2.256	2.262	2.319	2.257	2.301	2.329	2.272	2.336	2.333		
	Ebn at Noon		831	867	902	887	881	892	858	875	866	823	813	799		
	Edh at Noon		96	110	118	135	136	128	136	127	118	114	96	92		
(44) All-Sky Solar Radiation	RadAvg		2.61	3.61	5.08	6.43	7.54	8.45	7.99	7.43	6.24	4.57	3.19	2.39		
	RadStd		0.29	0.39	0.39	0.40	0.42	0.32	0.33	0.23	0.17	0.25	0.16	0.27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	+0.51	N/A	-1.13	+0.46	N/A	N/A	N/A	-123	+173	+68

Nomenclature: See separate page